

Multi-parameter models.

$$(\theta_1, \theta_2) \quad \pi(\theta_1, \theta_2) \quad \downarrow \text{prior}$$

$$x_1, \dots, x_n \sim_{\theta_1, \theta_2} f(x; \theta_1, \theta_2)$$

e.g. $\theta_1 = \mu, \theta_2 = \sigma^2$.

$$x_1, \dots, x_n \sim_{\mu, \sigma^2} N(\mu, \sigma^2)$$

$$\pi(\mu, \sigma^2) \checkmark$$

$$\pi(\theta_1, \theta_2 | D) \propto \pi(\theta_1, \theta_2) \pi(D | \theta_1, \theta_2)$$

$\mu, \sigma^2 \rightarrow$ nuisance parameters.

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D) d\theta_2$$

$$(X, Y) \quad x^{(1)}, y^{(1)} \\ x^{(2)}, y^{(2)} \\ \vdots \\ x^{(n)}, y^{(n)}$$

$$\pi(X, Y) = \pi(X | Y) \pi(Y)$$

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D) d\theta_2$$

$$= \int \pi(\theta_1 | \theta_2, D) p(\theta_2 | D) d\theta_2$$

$$\pi(\theta_1 | \theta_2, D) \quad \pi(\theta_2 | D)$$

Single-parameter inference.

$$\pi(\theta_1 | D)$$

Monte-Carlo integration

$$\theta_1^{(1)}, \theta_2^{(1)} \\ \vdots \\ \theta_1^{(B)}, \theta_2^{(B)}$$

Marginal of $\theta_1 | D$ can be obtained by ignoring the samples of θ_2

$$\theta_2 \rightarrow (M_1, M_2) \rightarrow \pi(\theta_1 | M_1, D)$$

$$\pi(\theta_2 | D) \checkmark$$

Suppose you want to estimate μ & σ^2 in $N(\mu, \sigma^2)$ model.
for known value of σ^2 you know how to estimate.
and you also know $p(\sigma^2 | D)$

$$\pi(\mu | D) = \int \pi(\mu | \sigma^2, D) \pi(\sigma^2 | D) d\sigma^2$$

$\sigma^2(\text{known})$ $\rightarrow$ this needs to be calculated

Generating sample from $\pi(\mu | D)$ can be done in 2-steps:

Step 1: Generate samples from $\sigma^2 | D$ (call it $\sigma^{2(s)}$)

Step 2: $\mu | \sigma^{2(s)}, D$

Gibbs sampling:

$$\sigma^2 | \mu, D \rightarrow \sigma^{(s)}$$

$$\mu | \sigma^2, D \rightarrow \mu^{(s)}$$

Conditional in Gibbs sampler.

Dependence.

$$\sigma^{(s+1)} | \mu^{(s)}, D$$

$$\mu^{(s+1)} | \sigma^{(s+1)}, D$$

Go to step 1 & Iterate

$$(X, Y) \quad \pi(X) \rightarrow \text{Generate from marginal dist}^n \text{ of } X$$

Scaled-inverted- χ^2 distribution:

$$f(x; \nu, \tau^2) = \tau^\nu \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \left(\frac{\nu}{x} \right)^{\frac{\nu}{2}+1} e^{-\frac{\nu \tau^2}{2x}}, \quad x > 0$$

& then $\pi(Y | X) \rightarrow$ condition distribution

Repeat

χ^2 normal unknown σ^2

Scaled-inv- $\chi^2(v, z^2)$. v is degrees of freedom.
 z^2 is scale

known μ .

If $X \sim \text{Scaled-inv-}\chi^2(v, z^2)$.

then a) $\frac{vz^2}{X} \sim \chi^2(v)$ b) $X \sim \text{inv-gamma}(\frac{v}{2}, \frac{vz^2}{2})$

c) $kX \sim \text{Scaled-inv-}\chi^2(v, kz^2)$

$\left[\frac{vz^2 \cdot k}{kX} \sim \chi^2(v) \Rightarrow kX \sim \text{Scaled-inv-}\chi^2(v, kz^2) \right]$

Let $X_1, \dots, X_n \text{ iid } N(\mu, \sigma^2)$ μ is known.

Let $\sigma^2 \sim \text{Scaled-inverse-}\chi^2(v_0, \sigma_0^2)$.

Then, $\sigma^2 | D \sim \text{Scaled-inverse-}\chi^2(v, s^2)$

where $v = v_0 + n$, $s^2 = \frac{v_0 \sigma_0^2 + \sum (X_i - \mu)^2}{v_0 + n}$

$\sigma^2 \sim \text{Scaled-inverse-}\chi^2(v_0, \sigma_0^2)$

$Y | \sigma^2 \sim N(\mu, \sigma^2)$ μ is known

Then, find the marginal distⁿ of Y .

$Y \sim t_{v_0}(\mu, \sigma_0^2)$

DO IT

2.
 $\sqrt{2}$

$p(y) = \int p(y | \sigma^2) \pi(\sigma^2) d\sigma^2$
 $y | \sigma^2 \sim N(\mu, \sigma^2)$

$\sigma^2 \rightarrow \text{known}$
 $\mu \rightarrow \text{unknown \& } \mu \text{ follows normal}$

Marginal of Y is normal.

$\sigma^2 | D \sim \text{Scaled-inv-}\chi^2(v_n, s_n^2)$

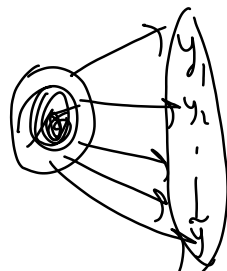
$p(\tilde{y} | D)$ (Posterior predictive distribution).

$y_1, \dots, y_n \text{ iid } N(\mu, \sigma^2)$ μ known.

$\sigma^2 \sim \text{Scaled-inv-}\chi^2(v_0, \sigma_0^2)$

$\sigma^2 | D \sim \text{Scaled-inv-}\chi^2(v_n, s_n^2)$

$y | \sigma, D$



$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$

Find $p(\mu, \sigma^2 | y)$

$y_1, \dots, y_n \text{ iid } N(\mu, \sigma^2)$

$p(\sigma^2 | y)$

Scaled-inverse- χ^2

Unknown Mean & Unknown Variance.

Assume prior as

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$$

$$p(\mu, \sigma^2 | y)$$

$$y_1, y_2, \dots, y_n \sim N(\mu, \sigma^2)$$

$$\underline{y}$$

Independent Jeffreys prior for μ & σ^2

$$p(\theta_1, \theta_2 | y)$$

$$\propto p(\theta_1 | \theta_2, y) p(\theta_2 | y)$$

When variance is known you can do this calculation.

$$p(\mu, \sigma^2 | y) \propto \sigma^{-n-2} \exp\left\{-\frac{\sum (y_i - \mu)^2}{2\sigma^2}\right\}$$

$$p(\mu | \sigma^2, y)$$

$$p(\sigma^2 | y)$$

$$p(\mu | \sigma^2, y)$$

When $\mu \sim N(\mu_0, \tau_0^2)$ & σ^2 is known.

Then, $p(\mu | \sigma^2, y) \sim N(\mu_1, \tau_1^2)$ where $\mu_1 = \left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{n}{\sigma^2}\right)\bar{y}$

$p(\mu) \propto 1$ can be obtained by taking $\tau_0 \rightarrow \infty$ in (A) & $\tau_1^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau_0^2}}$

$$p(\mu | \sigma^2, y) \sim N(\bar{y}, \frac{\sigma^2}{n}) \quad \text{--- (R1)}$$

$$p(\sigma^2 | y) \propto \int \sigma^{-n-2} \exp\left\{-\frac{1}{2\sigma^2} \left\{ \sum (y_i - \mu)^2 + n(\bar{y} - \mu)^2 \right\} \right\} d\mu$$

$$\propto \sigma^{-\frac{(n+1)}{2}} \exp\left\{-\frac{(n-1)s^2}{2\sigma^2}\right\} \text{ where } s^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$$

$$\Rightarrow \sigma^2 | y \sim \text{Scaled-Inverse-}\chi^2(n-1, s^2) \quad \text{--- (R2)}$$

Joint posterior

$$p(\mu, \sigma^2 | y) \propto \text{R1} \times \text{R2}$$

Sample from joint posterior.

Sample from $\sigma^2 | y$ &

... given the sample of σ^2

$$p(\mu | \sigma^2, y)$$

$$p(\sigma^2 | y) \times p(\mu | \sigma^2, y)$$

then sample $p(\mu|\sigma^2, n)$... Classical inference.

$R2 \Rightarrow \left(\frac{(n-1)s^2}{\sigma^2} \mid y \right) \sim \chi^2_{(n-1)} \leftrightarrow \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$

Pivotal ~~classical~~ quantity?

$N(\mu, 1)$ $N(0, 1) + \mu$

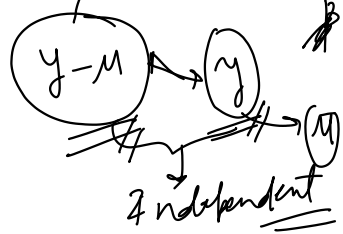
location family:

$p(y-\mu)$ is independent of μ .

pdf of $(y-\mu) \mid \mu$ is independent of μ . y is generated from a location family.

$\Rightarrow u = y - \mu$ and let $g(u)$ is the pdf of u .

(kind of) Ancillary for μ



Then. $p(y|\mu) = g(u) \left| \frac{du}{d\mu} \right| = g(u) = g(y - \mu)$

$p(y - \mu | y) = g(u)$

$u = y - \mu$

$p(u|y) = g(u)$
 $p(\mu|y) = g(u) \left| \frac{du}{d\mu} \right| = g(u)$

Reference prior

Pivotal quantity

$p\left(\frac{y}{\theta} \mid \theta\right)$ is a function that is free of θ

e.g. $2 \sim \text{Exp}(1)$ & then $y = 2\theta$

$\frac{y}{\theta}$ is a pivotal quantity here & θ is called a scale parameter.

$u = \frac{y}{\theta}$ $p\left(\frac{y}{\theta} \mid \theta\right)$ should be free of θ
 $u = \frac{y}{\theta}$

$$\Rightarrow (p(u|\theta) = \frac{1}{\theta} \quad (u \equiv y)$$

$$p(y|\theta) = g(u) \left| \frac{du}{dy} \right|$$

$$= \frac{g(u)}{\theta} = \frac{g(y/\theta)}{\theta}$$

$$\left(\frac{y}{\theta} \right) \quad (y)$$

distⁿ of y depend on θ & $\frac{y}{\theta}$ doesn't depend on θ .

$\Rightarrow \left(\frac{y}{\theta} \right)$ & y should be independent.

$$p\left(\frac{y}{\theta} | y\right) = \underline{g(u)}$$

$$u = \frac{y}{\theta} \Rightarrow \theta = \frac{y}{u} \Rightarrow \left| \frac{du}{d\theta} \right| \propto \frac{1}{\theta^2}$$

$$p(\theta|y) = p(u|y) \left| \frac{du}{d\theta} \right| = \frac{g(u)}{\theta^2}$$

$$p(\theta|y) \propto \pi(\theta) p(y|\theta)$$

$$\Rightarrow \frac{g(u)}{\theta^2} \propto \frac{\pi(\theta)}{\theta} g(u) \Rightarrow \boxed{\pi(\theta) \propto \frac{1}{\theta}}$$

Reference prior

Conjugate prior for unknown mean μ & unknown variance σ^2 .

$$p(y| \mu, \sigma^2) = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp \left\{ -\frac{\sum (y_i - \mu)^2}{2\sigma^2} \right\}$$

$$p(y|\mu, \sigma^2) \propto \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left\{ -\frac{\sum (y_i - \mu)^2}{2\sigma^2} \right\}$$

$$= (2\pi)^{-n/2} \exp \left\{ -\frac{\sum (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\}$$

$$= \underbrace{\left(\sigma^2 \right)^{-n/2} \exp \left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\}}_{\text{Scaled-Inverse-}\chi^2} \underbrace{\exp \left\{ -n \frac{(\bar{y} - \mu)^2}{2\sigma^2} \right\}}_{(\mu | \sigma^2) \sim N(\underline{\quad})}$$

Conjugate prior $\Rightarrow p(\mu, \sigma^2) \propto p(\sigma^2) p(\mu | \sigma^2)$

where $p(\sigma^2)$ is Scaled-Inverse- χ^2 and $\mu | \sigma^2 \sim N(\underline{\quad})$

$$\begin{aligned} \mu | \sigma^2 &\sim N(\underline{\mu_0}, \underline{\frac{\sigma_0^2}{k_0}}) \\ \sigma^2 &\sim \text{Inv-}\chi^2(\underline{\nu_0}, \underline{\sigma_0^2}) \end{aligned}$$

~~So far~~ $\Rightarrow p(\mu, \sigma^2) = N(\underline{\mu_0}, \underline{\frac{\sigma_0^2}{k_0}}) \times \text{Scaled-Inv-}\chi^2(\underline{\nu_0}, \underline{\sigma_0^2})$

$\Rightarrow \underline{p(\mu)} \quad \mu \sim t_{\nu_0}(\underline{\mu_0}, \underline{\frac{\sigma_0^2}{k_0}})$

$\underline{p(\mu, \sigma^2 | y)} = \underline{N\text{-inv-}\chi^2}(\mu, \sigma^2 | \underline{M_n}, \underline{\frac{\sigma_n^2}{K_n}})$

where $M_n = \frac{k_0}{k_0+n} \mu_0 + \frac{n}{k_0+n} \bar{y}$, $K_n = k_0 + n$, $\nu_0 = \nu_0 + n$, $\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + (n-1)s^2 + \frac{k_0 n}{k_0+n} [\bar{y} - \mu_0]^2$

$p(\mu | y)$

$p(\bar{y} | y)$

- ① Show ~~find~~ the distⁿ of $\bar{y}$ when prior $p(\mu)$ is $\frac{1}{\sigma^2}$
 $\mu | y \sim t_{n-1}(\bar{y}, \frac{s^2}{n})$
 ② Find the distⁿ of $\bar{y} | y$ ($\bar{y}$ is new obs.) using properties of normal & scaled-Inverse- χ^2 .

MAP & Credible Intervals

$$\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} L(\theta; x)$$

Data.
↓
parameter

Data is given, $L(\theta; x) = l(\theta|x)$

$\phi = h(\theta)$ h: one-one.

$$\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$$

MLE is invariant for 1-1 transformation
i.e. $\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$

Maximum A Posteriori Estimation (MAP)

$$\pi(\theta|x)$$

MAP is a point estimator of θ based on $\pi(\theta|x)$

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} \pi(\theta|x)$$

mode of $\pi(\theta|x)$

$X \sim N(\mu, \sigma^2)$ mode is μ

$$Y = e^X, X = \log Y$$

e^μ (is the mode?)

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$f_Y(y) =$$

Mode of Y will be $e^{\mu - \sigma^2} \neq e^\mu$

$$L(\theta; x) \Leftrightarrow p(x|\theta)$$

$L(\theta; x)$ (Likelihood function is not a prob. density fn.)

$$\int L(\theta; x) d\theta \neq 1 \quad \left[\int p(x|\theta) d\theta = 1 \right]$$

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} \pi(\theta|x)$$

$\pi(\theta|x)$ is a prob. density fn.
therefore MAP is not invariant to reparameterization.

$$= \underset{\theta}{\operatorname{argmax}} [L(\theta; x) \cdot \pi(\theta)]$$

Prior

e.g. $X_1, \dots, X_n \sim \exp(\beta)$
 $\pi(\beta) \propto 1$. Then $\hat{\beta}_{MAP} = \frac{\sum X_i}{n}$

$$pdf = \frac{1}{\beta} \exp(-\frac{x}{\beta})$$

$$\hat{\beta}_{MLE} = \frac{\sum X_i}{n}$$

$$\pi(\beta|x) \propto \frac{1}{\beta^n} \exp(-\frac{\sum X_i}{\beta}) \Rightarrow \pi(\phi|x)$$

$$\phi = \log \beta$$

then find $\hat{\phi}_{MAP}; \hat{\phi}_{MLE}$

$$\hat{\phi}_{MLE} = \log(\frac{\sum X_i}{n})$$

$$\log\left(\frac{\sum X_i}{n-1}\right)$$

Posterior mean $\pi(\phi|x)$
mode (MAP)

Credible interval.

$$P(L < \theta < U|x) = (1-\alpha)$$

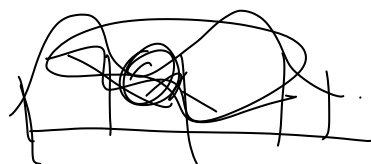
$$P(L < \theta < U) = 1-\alpha$$

e.g. $X|p \sim \text{Binomial}(n, p)$

$$p \sim \text{Beta}(a, b)$$

$$p|x \sim \text{Beta}(a+n, b+n-x)$$

$$\left(F^{-1}(0.025)_{\text{Beta}(a+n, b+n-x)}, F^{-1}(0.975)_{\text{Beta}(a+n, b+n-x)} \right)$$



$X_1, \dots, X_n \sim N(\mu, \sigma^2)$ σ known.

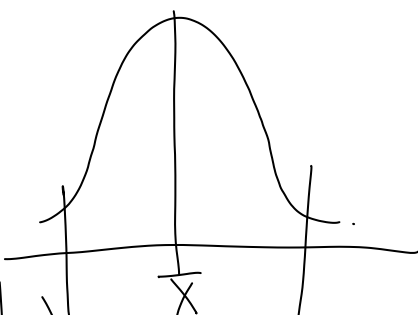
$$\pi(\mu) \propto 1$$

$$\text{Then, } \pi(\mu|x) \sim N(\bar{X}, \frac{\sigma^2}{n})$$

$$\mu|x_1, x_n \sim N(\bar{X}, \frac{\sigma^2}{n})$$

$$\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

$$P_x \left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \mid X \right) = 0.95$$



$\hat{\theta}_{MLE}$

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta) \sim N(0, I^{-1}(\theta))$$

$\left(\hat{\theta} - 1.96 \frac{\sigma}{\sqrt{n}}, \hat{\theta} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$ Asymptotic normality of MLE
Bernstein von Mises theorem $\left(\ln(\theta - \theta_0) \mid \mathcal{X} \sim N(0, \sigma_0^2) \right)$

- ① Paradox of classical inference.
- ② Why Bayesian. How to do Bayesian (prior, likelihood $\Rightarrow$ posterior.)
- ③ Prior $\rightarrow$ Conjugate prior (good for calculations).
 Some concepts of non-informative prior. (Jeffreys).
 prior, prior predictive distⁿ, posterior, posterior predictive distⁿ,
updating belief
 $\rightarrow$ Check at least all the cases mentioned in BS DS syllabus.
 Properties of scaled-inverse- χ^2 distⁿ
 $\rightarrow$ unknown mean unknown variance & Dirichlet Multinomial
 MAP & credible interval